AMAN CHAUDHARY

Computer Science and Engineering student

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SUMMARY

Computer Science and Engineering student with hands-on experience in **React.js**, **Node.js**, **Express.js**, and **MongoDB**. Passionate about building scalable full-stack web applications and developing **REST APIs**, while solving real-world problems through clean, efficient code and continuously learning modern technologies.

WORK EXPERIENCE

Web Developer | Internship

Jun 2026 - Jul 2026

InAmigos Foundation (IAF)

- Developed a responsive NGO awareness webpage using **HTML and CSS**.
- Designed **UI/UX feature proposals** in Figma for the organization's website.
- Researched and documented **volunteer opportunities and social initiatives**.

Technologies / Skills Used : *HTML, CSS, Agentic AI*

PROJECTS

PathFinder — Algorithm Visualization Tool | Github

- Built an interactive **React.js** visualization platform demonstrating **BFS, DFS, Dijkstra, and A*** pathfinding algorithms with step-by-step animations.
- Implemented a custom **Min-Heap Priority Queue** from scratch to efficiently support weighted pathfinding algorithms with optimized time complexity.
- Engineered **3 maze generation algorithms** (Recursive Division, Random, Spiral) with wall editing, weighted nodes, and drag-and-drop node placement.
- Delivered animation controls, dark mode toggle, and a performance metrics panel tracking visited nodes, path length, and execution time per run.

Technologies / Tools Used : *React.js, JavaScript, CSS3, DSA*

Secure Authentication & Authorization System | Github

- Developed a production-oriented REST API using **Node.js, Express.js, MongoDB, and JWT**, supporting registration, login, password recovery, email verification, and role-based access control.
- Implemented **short-lived access tokens and rotating refresh tokens** with HttpOnly cookies, SHA-256 hashing, and refresh-token reuse detection.
- Designed **server-side session management** with device-level session revocation and automatic invalidation after password changes or resets.
- Built centralized error handling, request validation, **bcrypt password hashing**, and MongoDB transactions for atomic security operations; tested API workflows using Postman.

Technologies / Tools Used : *Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, Express Validator, Nodemailer*

Library Management System | Github

- Developed a console-based library management system using **Java OOP**, encapsulation, collections, and layered separation of responsibilities.
- Implemented **book and member management**, including CRUD operations, unique ID validation, searching with `Optional`, and borrowing/return workflows.
- Designed a **custom exception hierarchy** and centralized exception handling for business-rule violations such as unavailable books, duplicate records, and borrowing-limit violations.
- Implemented member-book borrowing tracking with **borrowing records and borrowing limits**, maintaining consistency between book availability and member borrowing state.

Technologies / Tools Used : *java, OOP, Collections, Exception Handling, Optional*

SKILLS

Programming Languages :	Java, C++, C, Python, JavaScript, SQL
Frontend :	React.js, Bootstrap, HTML, CSS
Backend :	Node.js, Express.js, REST APIs, JWT, bcrypt
Databases :	MongoDB, MySQL
Tools & Platforms :	Git, GitHub, Postman

EDUCATIONS

KCC Institute of Technology and Management

2027

B.Tech in Computer Science & Engineering

CGPA : 6.4

Relevant Coursework: Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

Problem Solving (Basic) — HackerRank
JavaScript Developer — freeCodeCamp
Introduction to Generative AI — Simplilearn SkillUp (Google Cloud)